

# Aedes Control

## User Manual

*Simulation, intervention planning, and outcome comparison for Aedes aegypti mosquito control*

A guide for setting up, running the desktop application, and interpreting its results.

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## 1. What this application does

This application simulates the population of *Aedes aegypti* mosquitoes and the disease-transmission risk they pose at a location over time, and shows how a vector-control plan (larvicide, insecticide, and/or habitat removal) changes that outcome.

For a given location and time period, the application always runs two versions of the same simulation: one with no control measures applied, and one with your chosen control plan applied. It then shows you both curves side by side so you can see the impact.

The application offers two ways to build a control plan:

- **Optimize** — you decide how many times each treatment is applied and how strong each application is; the application searches for the timing that reduces transmission risk the most.
- **Fixed schedule** — you decide the exact calendar dates each treatment is applied on, and the application simply evaluates that plan.

## 2. Requirements: Python version and libraries

This application is written in Python 3 and relies on a small set of widely used, open-source libraries to do the numerical work and to draw the window itself. If someone has already set the application up for you, you can skip this section — it is provided so you know what is installed, and in case you ever need to set the application up on a new computer.

### Python version

Use **Python 3.10 or later** (the application has been tested on 3.10, 3.11, and 3.13). It relies on tkinter, which is part of the standard library on Windows and macOS Python installers; on Linux it may need to be installed separately (for example, `sudo apt install python3-tk` on Debian/Ubuntu).

### Required libraries

- **NumPy** — the array math used throughout the population model.
- **pandas** — reads and writes CSV files, and handles calendar dates and rolling averages of the daily results.
- **SciPy** — solves the mosquito population's differential equations (`scipy.integrate.solve_ivp`) and reads the built-in worldwide climate `.mat` files (`scipy.io.loadmat`).
- **Numba** — compiles the core population model to machine code just before it runs, so simulations and the Optimize search finish in a reasonable amount of time.
- **Matplotlib** — draws the population and transmission-risk charts and embeds them in the results dashboard.
- **tkinter** — builds the application window, forms, and buttons. It is included with most standard Python installations, so it usually does not need to be installed separately.

### Optional library

- **tkintermapview** — powers the “Select location on map” button in the Location section. The application runs fine without it; that one button will simply show a message asking you to install it if you click it before it is present.

### 3. Starting the application

The application is started by running the file named `Main.py`, using Python. If you are not sure how to start it on your computer, ask whoever set it up for you — typically this means double-clicking a shortcut, or running the following from a command prompt in the application's folder:

```
python Main.py
```

A window titled **Aedes Control** will open. The left side is where you configure a run; the right side is a dashboard that displays results once a run finishes (or once you load a previous run's results).

### 4. Choosing a mode: Optimize or Fixed schedule

At the top of the left panel are two tabs. Choose the one that matches what you want to do:

#### Optimize

Use this when you know how many times you want to apply each treatment, but you want the application to figure out the best timing to minimize transmission risk.

#### Fixed schedule

Use this when you already have specific calendar dates in mind — for example, dates matching a real-world spraying campaign — and you want to see what effect that specific plan has.

## 5. Setting up a run

Both tabs ask for the same three groups of information, described below.

### 5.1 Control measures

For each of the three treatments — Insecticide, Larvicide, and Habitat removal — you provide:

- **Duration (days)** — how many days a single application keeps working.
- **Efficiency (0 to 1)** — how effective a single application is, where 1 means fully effective.
- **Optimize tab: Applications** — how many times that treatment is applied during the simulated period.
- **Fixed schedule tab: Application dates** — the exact dates that treatment is applied on, separated by commas, in the form YYYY-MM-DD (for example 2024-06-01, 2024-07-15). You can type these directly, or click **From CSV...** to load them from a file — see Section 9.

### 5.2 Data and output

This section tells the application what climate conditions to simulate, and where to save the results.

**Temperature data** and **Precipitation data** — the application comes with ready-to-use climate data files (`temp_pastyearsav_py.mat` and `pre_pastyearsav_py.mat`), already filled in by default. These files contain temperature and rainfall averaged over recent years (for example, 2023–2024) for locations worldwide. If you use these default files, you do not need to prepare any climate data yourself — you only need to set the Location (Section 5.3) and the application will automatically read the conditions for that spot.

If you would rather supply your own climate record instead of the built-in average — for example, real data for a specific year at your site — you can replace either field with your own CSV file. See Section 8 for exactly what that file must look like.

**Results folder** — the folder where this run's results will be saved. Type a folder name or path, or click **Browse** to choose one. The folder is created automatically if it does not already exist.

### 5.3 Location

Enter the latitude and longitude of the site you want to simulate, or click **Select location on map** to choose a point visually. Location is only used when the climate inputs are the default map-grid files; if you supply your own climate CSV files, the location entered here is not used, because your file is already specific to one place.

## 6. Running the model

Once the fields above are filled in, click **Run optimization** or **Run fixed schedule** at the bottom of the panel. The application keeps working in the background while the run proceeds — this can take anywhere from under a minute to several minutes, depending on your settings. You can follow progress in the status bar at the bottom of the window and in the **Run log** tab on the right. When the run finishes, its results load into the dashboard automatically.

## 7. Reading your results

The dashboard on the right has three tabs, plus three summary numbers at the top:

- **Average population reduction** and **Peak risk reduction** — how much the control plan lowered mosquito numbers and transmission risk, on average and at the worst point.
- **Days modeled** — the length of the simulated period.

### Outcome charts

Two charts show mosquito population and transmission risk over time, each with a line for “before control” and a line for “after control.” You can scroll to zoom into a time range, use **Pan** or **Box zoom** to navigate, or use **Select date window** to drag across a specific period. **Reset view** returns to the full period.

### Treatment schedule

One card per treatment shows exactly which dates it was applied on, along with its duration and efficiency.

### Run log

A running log of what the application has done during this session, useful if something goes wrong and you need to see what happened.

You do not need to run a new simulation to see old results: click **Load results** at any time and choose a previous results folder, or click **Refresh** to reload the currently open one.

## 8. Using your own climate data

Instead of the built-in average climate files, you may provide your own daily temperature and/or precipitation record as a CSV file. This is useful if you have real measurements for a specific site and year. Your file must follow these rules:

- Two columns: a date column and a value column. The date column can have any header name (or none).
- Covers exactly one full calendar year, from January 1 through December 31 — 365 rows, or 366 rows in a leap year — with one row per day and no missing or repeated dates.
- **Temperature values must be in degrees Celsius.**
- **Precipitation values must be in meters per day.**
- If you provide both a temperature CSV and a precipitation CSV, they must cover the exact same year.

Unlike the built-in files, a CSV file is not looked up by location — it is assumed to already represent the single site you care about.

## 9. Loading treatment dates from a file

In the Fixed schedule tab, instead of typing application dates by hand, you can click **From CSV...** next to any treatment to load them from a file. That file should be a simple single-column list of dates, one per row, in the form YYYY-MM-DD (other common date formats are also accepted). A header row (such as the word “date”) is fine — it is skipped automatically. Once loaded, the dates appear in the text box, where you can still review or edit them by hand before running the model.

## 10. Common questions and messages

### **“All treatment dates must use the same year”**

This appears when you are using the built-in climate files (which represent one averaged, undated year) together with treatment dates that span more than one calendar year. Either keep all your treatment dates within a single year, or switch to a dated CSV climate file, which does allow multi-year context.

### **“Treatment dates must be between ...”**

A date you entered falls outside the period covered by your climate data. Check the dates against the year your climate file represents.

### **“Complete all required fields before running”**

One or more required boxes (duration, efficiency, location, data files, results folder) is empty.

### **The run finished but the dashboard could not load its output**

The run completed and its files were saved to the results folder, but something prevented the dashboard from displaying them (for example, an unexpected file format). You can still open the results folder directly, or try Load results again.

## 11. Reference: files saved by each run

Every run saves its results as plain text files inside the results folder you chose. You do not need these to use the application — the dashboard reads them for you automatically — but they are listed here in case you want to open them yourself in another program, such as a spreadsheet.

| File                                                                                                          | Contents                                                                                                                    |
|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| dates.txt                                                                                                     | The daily calendar for the simulated period.                                                                                |
| population_before.txt<br>population_after.txt                                                                 | Adult mosquito population, before and after control.                                                                        |
| risk_before.txt<br>risk_after.txt                                                                             | Transmission-risk index, before and after control.                                                                          |
| larvicide_application_dates.txt<br>insecticide_application_dates.txt<br>habitat_removal_application_dates.txt | The dates each treatment was actually applied on.                                                                           |
| run_metadata.json                                                                                             | A summary of the simulated period, the climate data used, and each treatment's duration, efficiency, and application count. |
| optimization_convergence.txt                                                                                  | Optimize mode only: a record of how the search for the best schedule progressed.                                            |

## 12. Reference: backend Python files

The application is made up of several Python files that work together. You do not need to read any of this to use the application — it is provided for anyone curious about how the simulation works under the hood, or who needs to maintain or extend the code.

| File                                   | What it does                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>Main.py</code>                   | The application window itself, built with tkinter. Collects your inputs from the Optimize and Fixed schedule forms, runs a simulation in the background so the window stays responsive, and draws the resulting charts, treatment schedule, and run log in the results dashboard.                                                   |
| <code>Run_AEDES_AEGYPTI.py</code>      | Runs “Fixed schedule” mode. Builds the simulated calendar from the chosen climate data, places your treatment dates on it, solves the population model once without controls and once with them, and writes the before/after result files listed in Section 11.                                                                     |
| <code>Optimize_AEDES_AEGYPTI.py</code> | Runs “Optimize” mode. Repeatedly simulates the population and, using a gradient-based search over several hundred iterations, adjusts each treatment’s timing step by step until it converges on the schedule that most reduces transmission risk. Also writes <code>optimization_convergence.txt</code> , a record of that search. |
| <code>Dif_functions.py</code>          | Defines the mosquito population model as a system of differential equations (egg, larva, pupa, and adult life stages), together with the numerical routines <code>Optimize_AEDES_AEGYPTI.py</code> uses to work out how sensitive the outcome is to each candidate treatment date. Compiled with Numba so simulations run quickly.  |
| <code>MosquitoRates.py</code>          | Supplies the temperature-dependent biological rates — development and mortality rates for each life stage, adult mortality, egg-laying rate, and mosquito-to-human/human-to-mosquito transmission rates — that <code>Dif_functions.py</code> uses at each simulated day’s temperature.                                              |
| <code>LocationSeriesInput.py</code>    | Loads and checks the temperature and precipitation data for a run, either from the built-in worldwide climate .mat files (looked up by the location you enter) or from a CSV file you supply, and lines the two series up into one daily calendar for the simulation.                                                               |
| <code>cord_to_CRoFlanl.py</code>       | A small helper that converts a latitude/longitude pair into the row and column used to look up that location’s data inside the built-in worldwide climate .mat files.                                                                                                                                                               |